

# Workshop on Computational Models of Human Road User Behavior for Evaluation of Automated Vehicles

September 15, 2026, Naples, Stazione Marittima, Room Calipso

Abstracts of contributed talks and most recent schedule are available at <https://roadusermodels.github.io/>

| Session                               | Start | End   | Duration (min) | Item Type    | Speaker(s)                     | Affiliation                                | Presentation Title / Topic                                                                                                        |
|---------------------------------------|-------|-------|----------------|--------------|--------------------------------|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Session 1.<br>Chair: Arkady Zgonnikov | 8:30  | 8:40  | 10             | Logistics    | Arkady Zgonnikov               | TU Delft, Organizer                        | Welcome, Workshop Objectives & Agenda                                                                                             |
|                                       | 8:40  | 9:10  | 30             | Keynote      | Biagio Ciuffo                  | European Commission – JRC                  | Driving behaviour models to operationalize the international regulatory framework for AV safety approval                          |
|                                       | 9:10  | 9:40  | 30             | Keynote      | Hiroki Nakamura & Sou Kitajima | Japan Automobile Research Institute (JARI) | Defining Safety Boundaries for Automated Driving Systems: From real traffic data to computational human road-user models          |
| Session 2.<br>Chair: Jonas Bärgman    | 9:40  | 10:00 | 20             | Contributed  | Kevin Kefauver                 | VTTI                                       | Prudent Attentive Sober (PAS) Driver Definition in US National Crash Data and its Application to Human Behavioral Models          |
|                                       | 10:00 | 10:20 | 20             | Contributed  | Arturo Tejada Ruiz             | TNO                                        | Bending the (traffic) rules, competent driver style!                                                                              |
|                                       | 10:20 | 10:40 | 20             | Contributed  | Giovanni Albano                | Univ. of Naples Federico II                | Risk perception and braking reaction model of competent and careful human drivers in safety-critical cut-in scenarios             |
|                                       | 10:40 | 11:00 | 20             | Contributed  | Federico Scari                 | RDW                                        | A Scenario-Based Grading Framework for ADS/ADAS Consumer-Testing Campaigns Benchmarked Against a Probabilistic Human Driver Model |
| Break                                 | 11:00 | 11:30 | 30             | Coffee Break |                                |                                            |                                                                                                                                   |
| Session 3.<br>Chair: Gustav Markkula  | 11:30 | 12:00 | 30             | Keynote      | Jonas Bärgman                  | Chalmers Univ. of Technology               | Representativeness in ADS safety impact assessment (and training): baseline and traffic-agent challenges                          |
|                                       | 12:00 | 12:30 | 30             | Keynote      | Manel Hammouda                 | BMW Group                                  | Cognition and Reward: A Dual Perspective on Driver Behavior Modeling                                                              |
|                                       | 12:30 | 12:50 | 20             | Contributed  | Yueyang Wang                   | Univ. of Leeds                             | Task-oriented scoring of driver behaviour models for automated driving systems evaluation                                         |
|                                       | 12:50 | 12:53 | 3              | Teaser talk  | Malin Svärd                    | Volvo Cars                                 | Quantification of Human Acceleration Behavior across Speeds Using Real-World Driving Data                                         |
|                                       | 12:53 | 12:56 | 3              | Teaser talk  | Konstantinos Mattas            | European Commission – JRC                  | Combining Evasive and Braking Reactions for Safety Reference Models in Automated Vehicles                                         |
|                                       | 12:56 | 12:59 | 3              | Teaser talk  | Samir Mohammad                 | TU Delft                                   | General-purpose LLMs as Models of Human Driver Behavior: The Case of Simplified Merging                                           |
| Break                                 | 13:00 | 14:00 | 60             | Lunch Break  |                                |                                            |                                                                                                                                   |
| Session 4.<br>Chair: Luigi Pariota    | 14:00 | 14:30 | 30             | Keynote      | Gustav Markkula                | Univ. of Leeds                             | Mechanistic, machine-learned, and hybrid modelling of road user behaviour                                                         |
|                                       | 14:30 | 14:50 | 30             | Keynote      | Johan Engström                 | Waymo                                      | SAE J3330: Toward a shared language for driving behavior reference models                                                         |
| Break                                 | 15:00 | 15:15 | 15             | Break        |                                |                                            |                                                                                                                                   |
| Session 5.<br>Chair: Johan Engström   | 15:15 | 15:35 | 20             | Contributed  | Arash Tavakoli                 | Villanova Univ.                            | Driver World Models: Unified Representations of Drivers in Semi-Automated Driving                                                 |
|                                       | 15:35 | 15:55 | 20             | Contributed  | Christian Rössert              | cogniBIT GmbH                              | Beyond benchmark metrics: a cognitive driver model as a behavioral reference for AV evaluation                                    |
|                                       | 15:55 | 16:15 | 20             | Contributed  | Saeed Rahmani                  | TU Delft                                   | Drive with Style: Style-Conditioned Self-Play for Behaviorally Diverse Traffic Simulation                                         |
|                                       | 16:15 | 16:35 | 20             | Contributed  | Julian Frederik Schumann       | TU Delft                                   | Active Inference as a model of human driving behavior                                                                             |
| Break                                 | 16:35 | 16:45 | 10             | Break        |                                |                                            |                                                                                                                                   |
| Session 6.<br>Chair: Shu-Yuan Liu     | 16:45 | 17:35 | 50             | Discussion   | Shu-Yuan Liu, Arkady Zgonnikov | Moderated by Organizers                    | Interactive Breakout Groups & Plenary Synthesis                                                                                   |
|                                       | 17:35 | 17:45 | 10             | Logistics    | Arkady Zgonnikov               | Organizing Committee                       | Closing Remarks                                                                                                                   |